Meshaal Mouawad
Mississippi State University, Electrical and Computer Engineering Department,
*This is a summary report of my research assignments I was given by my advisor during my PhD program*

# SUMMARY REPORT: FOR RESEARCH ASSIGNMENTS OF MY PHD

**WEEK 17**

July 10, 2021

# Voronoi Diagram Project

3. **Environmental Exploration**: Construction of TWO kinds of Voronoi Diagram maps: Voronoi Diagram, and Generalized Voronoi Diagram (GVD) as follows. We do not need to implement and program VD, and GVD. There are some MTALB functions/commands for VD available. GVD might be available in MTALB as well, Or we may find online with MTALB file exchange, github, Sci-hub (https://en.wikipedia.org/wiki/Sci-Hub), or other resources.

## Start Generating VD from the MATLAB code:

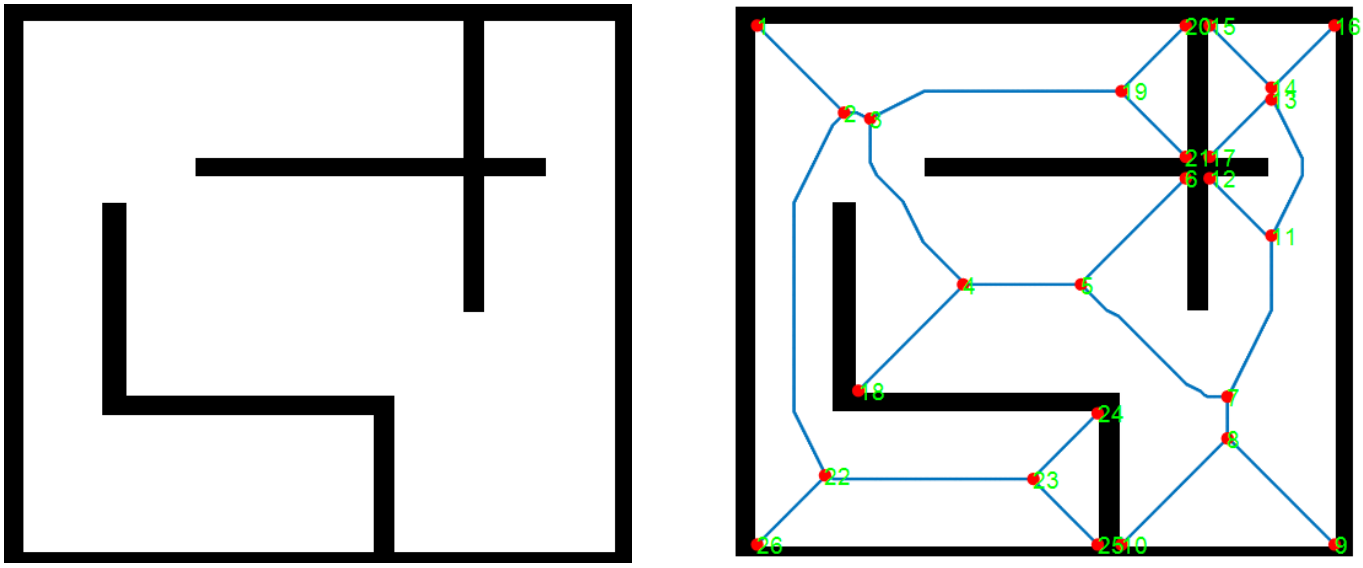



**Figure. 2 the result from the matlab code**

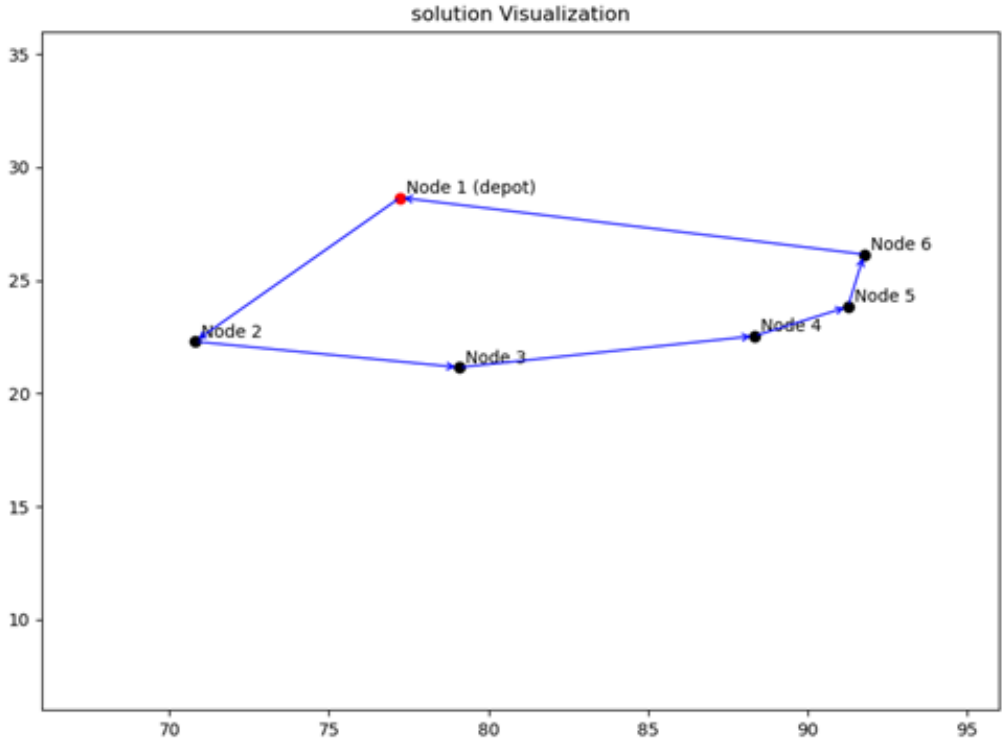
4. **Multi-goal path planning:** Once the construction of Voronoi Diagram (GVD) maps is accomplished, some algorithms for TSP (Travelling salesman problem) is applied to connect some goals/waypoints that we need to explore, rescue, and reach. Running your TSP algorithm to connect some points such as green stars:

I used another Python code for TSP and successfully run it! **Gorubi** Module was used for optimization. The output for the optimal path is:

```
[(0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 0)]

Time taken by MTZ formulation =  1.2135167121887207
```

**The solution visualization is:**



5. **Local navigator using DWA (Dynamic Window Approach):** ) method to walk a mobile robot from starting point to goal along the edges of built GVD. DWA to navigate the robot to connect multiple goals depicted by the green stars.

   **I am waiting the code you told me to send it to me**

# After that, the next step will be the Mapping

# Conclusion

**VD**
Generate VD
**TSP:**
to connect multiple points, and minimize traveling distant.

**DWA:**
Path planning and local navigation. To walk the robot from **start** point to the **goal** along the edge of GVD.

**Mapping:**